

Luca Tomei

Software Engineer · IT Development Advisor

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· github.com/LucaTomei



Professional Summary

Software Engineer (MSc Engineering in Computer Science, Sapienza University of Rome) working on enterprise systems since 2021: full-stack development with .NET 8, Angular and SQL Server, ERP integration on Infor M3 (on-premise and cloud) through ION, MEC and ESB middleware, microservices, REST APIs and low-code delivery on OutSystems. Currently IT Development Advisor at Yokohama TWS, defining development standards, governing the CI/CD and containerisation strategy and coordinating in-house teams, external consultants and 3PL partners. Production-grade mindset and end-to-end ownership, from requirements analysis to deployment and support. Outside work I build open-source automation, embedded and NFC projects.

Technical Skills

Languages: C#, Python, TypeScript, JavaScript, Java, SQL, PHP, C/C++, Scala, Swift, Kotlin, PowerShell

Backend & APIs: .NET 8, Laravel, Node.js, REST APIs, Microservices, RabbitMQ, XSD/XML, JSON

Frontend & Mobile: Angular, Vue.js, React, React Native, SwiftUI

ERP & Integration: Infor M3 (on-premise and cloud), Infor ION, MEC, ESB, 3PL integrations, OutSystems

Databases & Data: SQL Server, MySQL, PostgreSQL, MongoDB, SQLite, Databricks

DevOps & Cloud: Azure DevOps, GitHub Actions, CI/CD, Docker, Linux, IIS, Proxmox, Cloudflare, Azure

IoT & Embedded: ESP32, MQTT, NFC, Flipper Zero, Home Assistant, Node-RED, firmware reverse engineering

Design & CAD: Autodesk Fusion 360, AutoCAD, 3D printing

Work Experience

IT Development Advisor

February 2026 – Present

Yokohama TWS (formerly Trelleborg Wheel Systems) – Tivoli (Rome), Italy

- Define software development strategy and best practices, governing the adoption of microservices, Azure DevOps CI/CD pipelines and Docker-based deployments
- Provide technical leadership on code quality and performance across the .NET 8, Angular and SQL Server stacks
- Design and govern enterprise API orchestration between microservices and legacy systems through Infor ION, MEC and ESB middleware
- Coordinate the Infor M3 cloud migration and the rollout of OutSystems low-code applications aligned with business requirements
- Manage outsourced development projects, aligning external consultants and 3PL integration partners with company standards; turn business requirements into scalable, documented technical specifications

IT Development Specialist

May 2023 – February 2026

- Led technical analysis, development and implementation of custom solutions on Infor M3 ERP (on-premise) and the OutSystems low-code platform
- Developed REST APIs and C#/.NET backend microservices integrated with Angular enterprise dashboards and SQL Server databases
- Designed and maintained enterprise integration flows (MEC, ESB, 3PL) with XSD/XML schema design and data mapping; troubleshoot inbound/outbound documents with external logistics partners
- Built and maintained Azure DevOps CI/CD pipelines for automated deployment of frontend, backend and database components
- Performed code reviews and root cause analysis; maintained technical documentation and runbooks

Stack: .NET 8, C#, Angular, TypeScript, SQL Server, MySQL, PHP/Laravel, Vue.js, RabbitMQ, Infor M3, Infor ION, MEC, ESB, OutSystems, Azure DevOps, Docker, IIS

ERP Developer

November 2021 – May 2023

Trelleborg Wheel Systems – Tivoli (Rome), Italy

- Managed the technical side of the Infor M3 ERP: Business Engine, interfaces, layouts and reports; integrated the ERP with the other business systems
- Analysed functional documentation, wrote technical documentation, fixed bugs and kept ERP versions aligned with vendor releases
- Supported the business on technical ERP issues and worked with vendor consultants on complex implementations

Teaching Assistant, Functional and Parallel Programming

October 2021 – December 2021

Sapienza University of Rome – Rome, Italy

- Tutored the online section (about 100 students) of the Scala-based course held by Prof. Camil Demetrescu: labs, programming assignments, quizzes and exams

Home Automation and IT Technician (freelance)

November 2015 – October 2021

Self-employed – Rome, Italy

- Installed and maintained home automation and control systems for private homes and commercial spaces; built databases for business data management and provided technical consulting

Education

Master's Degree in Engineering in Computer Science – 110/110

January 2020 – October 2021

Sapienza University of Rome – Rome, Italy

- Thesis: *Design and proof-of-concept of the integration of an electric scooter with a wheelchair: from the mechanical design to the firmware upgrade* – custom firmware with adaptive acceleration, Python client/GUI for firmware builds, 3D-designed bracket

Bachelor's Degree in Computer and Automation Engineering – 99/110

2016 – 2019

Sapienza University of Rome – Rome, Italy

- Thesis: *Mapping Management of an Ontology-based Data Access System* – Java and React web application for browser-side ontology mapping and querying

Scientific High School Diploma – Liceo Scientifico Ettore Majorana, Guidonia (Rome)

2009 – 2014

Projects

Electric Scooter Wheelchair – github.com/LucaTomei/Electric-Scooter-Wheelchair

2020 – 2021

- Working prototype coupling an electric scooter to a wheelchair: CAD-designed, 3D-printed bracket and reverse-engineered firmware with acceleration and braking tuned for the new load (MSc thesis) (Python, firmware, Autodesk Fusion 360, 3D printing)

OPBG Telegram Booking Bot

Private project

- Invite-only multi-user bot automating the booking portal of the Bambino Gesù children's hospital: search by treatment or doctor, availability monitoring with notification or auto-booking, appointment management; separate bot and worker processes, idempotent booking protocol, admin panel (Python, aiogram, SQLAlchemy, MySQL, Docker)

Flipper Mi Band NFC Writer – github.com/LucaTomei/flipper-miband-nfc-writer

2025

- Flipper Zero app to read, write, emulate and verify NFC data on Xiaomi Mi Band devices: automatic backups, sector-level progress, hex diff viewer, persistent settings and exportable logs (C, Flipper Zero, NFC, MIFARE Classic)

Telegram Audio Transcriber – github.com/LucaTomei/Telegram-Audio-Transcriber

2026

- Self-hosted bot transcribing voice messages and audio files locally with Whisper models; optional text cleanup via any OpenAI-compatible endpoint, whitelist and admin roles, no audio persistence; Docker Compose deployment from CI-built images (Python, Faster-Whisper, ffmpeg, SQLAlchemy, Docker, GitHub Actions, GHCR)

LazioHealthMonitorBot – github.com/LucaTomei/LazioHealthMonitorBot

2025 – Present

- Multi-user bot polling the Lazio regional health service every five minutes: real-time notifications, date filters, hospital blacklist, direct or automatic booking of the first free slot, PDF confirmations; CI validation and container images on every release (Python, Telegram Bot API, REST, Docker, GitHub Actions, GHCR)

Carburanti Bot – github.com/LucaTomei/Carburanti-Bot

2026

- Real-time fuel price bot for Italy: favourite station, nearby-station and best-price search from a shared location, daily notifications, keyboard-driven UI and admin panel (Python, Docker, Portainer, MIMIT/Osservaprezzi open data)

Estrai Allegati – github.com/LucaTomei/estrai-allegati

2026

- Native macOS app extracting embedded files from Word, Excel and PowerPoint documents: dependency-free ZIP, OLE2/Compound File and EMF parsers written from scratch, universal binary, automated DMG releases (Swift, SwiftUI, macOS, GitHub Actions)

Computer Scientists Retrieval – github.com/LucaTomei/Computer_Scientists

2020

- Web Information Retrieval project ranking computer scientists by influence with PageRank and HITS over a Wikipedia/DBpedia graph, with category classification (Python, NetworkX, SPARQL/DBpedia, BeautifulSoup)

Other open source

2019 – Present

- **MAXG30CFW** – Ninebot Max G30 firmware builder (Python, reverse engineering) · **telegram-ha-monitor** – multi-server monitoring bot with leader election and RBAC · **Esp32-Cam-PIR-Telegram** – motion-triggered camera controlled via Telegram (C++) · **GL.iNet VPN Switcher** (OpenWrt) · **Hoodie** (iOS) and **Terminal Watch Face** (watchOS) in Swift · **LightTextEditor** (C) · **FlyingBear Ghost 5** (3D printing)

Certifications

OutSystems Associate Reactive Developer – OutSystems, 2024

Software Architecture Fundamentals; Artificial Intelligence Fundamentals: Machine Learning – Birdify – LMSAFP, 2024

Introducing Semantic Kernel: Building AI-Based Apps – LinkedIn Learning, 2024

Introduction to iOS Mobile Application Development – Meta, Coursera, 2023

Academy Accreditation – Databricks Lakehouse Fundamentals – Databricks, 2022

Introduction to Cloud Computing; Introduction to Cybersecurity Tools & Cyber Attacks; Introduction to Data Analytics – IBM, Coursera, 2022

Foundations of User Experience (UX) Design (2022); Crash Course on Python (2021) – Google, Coursera

REST API (Intermediate); SQL (Basic, Intermediate); Java, Python, React (Basic) – HackerRank, 2021

Languages

Italian – Native · English – B2 (CEFR) · French – A2 (CEFR)

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